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Blood vessel anastomosis set

Abstract

A blood vessel anastomosis set for anastomosing a blood vessel cut of a blood vessel is provided. The blood vessel anastomosis set includes: a blood vessel anastomosis device having a plurality of anastomosis needles and a plurality of anastomosis needle holes; and a blood vessel fixation device including: a blood vessel widening portion for widening the blood vessel cut; and a blood vessel fixation portion for fixing the blood vessel cut to the blood vessel anastomosis device, wherein the blood vessel anastomosis set has a protection mechanism whereby the blood vessel fixation device is not in contact with the plurality of anastomosis needles when the blood vessel cut is fixed to the blood vessel anastomosis device.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present disclosure relates to blood vessel anastomosis sets and, more particularly, to a blood vessel anastomosis set for anastomosing a blood vessel cut.

Description of the Prior Art

(2) Conventional surgical procedures often entail anastomosing a blood vessel which has been cut apart. For instance, a conventional procedure of reconstruction after a mastectomy entails anastomosing severed blood vessels. However, the lengthy process of suturing blood vessels is excruciating and thus risky to patients.

(3) In view of the aforesaid drawback of the prior art, it is necessary to provide a blood vessel anastomosis set conducive to speeding up a surgical procedure of anastomosing a blood vessel cut.

SUMMARY OF THE INVENTION

(4) In view of the aforesaid drawback of the prior art, it is an objective of the present disclosure to provide a blood vessel anastomosis set conducive to speeding up a surgical procedure of anastomosing a blood vessel cut.

(5) In order to achieve the above and other objectives, the present disclosure provides a blood vessel anastomosis set for anastomosing a blood vessel cut of a blood vessel. The blood vessel anastomosis set comprises: a blood vessel anastomosis device having a plurality of anastomosis needles and a plurality of anastomosis needle holes; and a blood vessel fixation device, comprising: a first blood vessel widening portion for widening the blood vessel cut; and a blood vessel fixation portion for fixing the blood vessel cut to the blood vessel anastomosis device, wherein the blood vessel anastomosis set has a protection mechanism whereby the blood vessel fixation device is not in contact with the plurality of anastomosis needles when the blood vessel cut is fixed to the blood vessel anastomosis device.

(6) In a preferred embodiment of the present disclosure, the blood vessel anastomosis device is of a ring-shaped structure and has an inner hole which the blood vessel is fitted in.

(7) In a preferred embodiment of the present disclosure, the blood vessel anastomosis device is connected through the plurality of anastomosis needles and the plurality of anastomosis needle holes to a second blood vessel anastomosis device fixed to another blood vessel cut.

(8) In a preferred embodiment of the present disclosure, the blood vessel anastomosis device comprises a front-end anastomosis portion and a rear-end anastomosis portion, wherein the plurality of anastomosis needles are disposed at the rear-end anastomosis portion and penetrate the front-end anastomosis portion, thereby allowing the rear-end anastomosis portion to be movably connected to the front-end anastomosis portion by the anastomosis needles.

(9) In a preferred embodiment of the present disclosure, the protection mechanism prevents the plurality of anastomosis needles from protruding from the front-end anastomosis portion when the blood vessel fixation portion fixes the blood vessel cut to the blood vessel anastomosis device.

(10) In a preferred embodiment of the present disclosure, the blood vessel fixation portion has a receiving portion, and the protection mechanism enables the plurality of anastomosis needles to be received in the receiving portion when the blood vessel fixation portion fixes the blood vessel cut to the blood vessel anastomosis device.

(11) In a preferred embodiment of the present disclosure, the blood vessel fixation device further comprises a second blood vessel widening portion movably received in the blood vessel fixation device, wherein the second blood vessel widening portion widens the blood vessel from inside.

(12) In a preferred embodiment of the present disclosure, the blood vessel fixation device further comprises a retracting portion whereby the second blood vessel widening portion restores a retracted state thereof early as soon as the second blood vessel widening portion is retracted into the blood vessel fixation device.

(13) In a preferred embodiment of the present disclosure, the front end of the first blood vessel

widening portion and/or the second blood vessel widening portion is provided with one of an inclined surface, a concave arc surface and an outward concave arc surface to gradually widening the blood vessel cut.
(14) In a preferred embodiment of the present disclosure, the blood vessel anastomosis device has a snap-engagement portion, and the blood vessel fixation portion is separably disposed at the blood vessel fixation device, wherein, as soon as the blood vessel fixation portion fixes the blood vessel cut to the blood vessel anastomosis device, the blood vessel fixation portion is snap-engaged with the snap-engagement portion, and the blood vessel cut is clamped between the blood vessel fixation portion and the blood vessel anastomosis device.
(15) In order to achieve the above and other objectives, the present disclosure further provides a blood vessel anastomosis set for anastomosing a blood vessel cut of a blood vessel. The blood vessel anastomosis set comprises: a blood vessel anastomosis device; and a blood vessel fixation device for widening the blood vessel cut to allow the blood vessel cut to be fitted to the blood vessel anastomosis device, wherein the blood vessel anastomosis device comprises: a front-end anastomosis portion; and a rear-end anastomosis portion having a plurality of anastomosis needles, wherein the plurality of anastomosis needles penetrate the front-end anastomosis portion, such that the rear-end anastomosis portion is movably connected to the front-end anastomosis portion through the anastomosis needles.
(16) In a preferred embodiment of the present disclosure, the blood vessel anastomosis set is connected through the plurality of anastomosis needles to a second blood vessel anastomosis device fixed to another blood vessel cut.
(17) In a preferred embodiment of the present disclosure, the plurality of anastomosis needles do not protrude from the front-end anastomosis portion when the blood vessel fixation device fixes the blood vessel cut to the blood vessel anastomosis device.
(18) In a preferred embodiment of the present disclosure, the blood vessel fixation device comprises a foldable blood vessel widening portion, and the blood vessel fixation device widens the blood vessel cut through the foldable blood vessel widening portion.
(19) The foregoing aspects and other aspects of the present disclosure are illustrated by non-restrictive specific embodiments, depicted by the accompanying drawings, and described below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A is a schematic view of how to operate a blood vessel anastomosis set according to a specific embodiment of the present disclosure.
(2) FIG. **1**B is a schematic view of how to operate the blood vessel anastomosis set according to a specific embodiment of the present disclosure.
(3) FIG. **1**C is a schematic view of how to operate the blood vessel anastomosis set according to a specific embodiment of the present disclosure.
(4) FIG. **1**D is a schematic view of how to operate the blood vessel anastomosis set according to a specific embodiment of the present disclosure.
(5) FIG. **1**E is a schematic view of how to operate the blood vessel anastomosis set according to a specific embodiment of the present disclosure.
(6) FIG. **1**F is a schematic view of how to operate the blood vessel anastomosis set according to a specific embodiment of the present disclosure.
(7) FIG. **1**G is a schematic view of how to operate the blood vessel anastomosis set according to a specific embodiment of the present disclosure.
(8) FIG. **2** is a schematic view of a blood vessel fixation device comprising a second blood vessel widening portion according to a specific embodiment of the present disclosure.

(9) FIG. **3** is a schematic view of a blood vessel anastomosis device according to a specific embodiment of the present disclosure.
(10) FIG. **4** is a schematic view of a blood vessel anastomosis device according to another specific embodiment of the present disclosure.
(11) FIG. **5**A is a schematic view of how to operate the blood vessel anastomosis set according to another specific embodiment of the present disclosure.
(12) FIG. **5**B is a schematic view of how to operate the blood vessel anastomosis set according to another specific embodiment of the present disclosure.
(13) FIG. **5**C is a schematic view of how to operate the blood vessel anastomosis set according to another specific embodiment of the present disclosure.
(14) FIG. **5**D is a schematic view of how to operate the blood vessel anastomosis set according to another specific embodiment of the present disclosure.
(15) FIG. **5**E is a schematic view of how to operate the blood vessel anastomosis set according to another specific embodiment of the present disclosure.
(16) FIG. **6** is a schematic view of a blood vessel fixation portion is snap-engaged with a blood vessel anastomosis device according to a specific embodiment of the present disclosure.
(17) FIG. **7**A is a schematic view of how to operate the blood vessel anastomosis set according to yet another specific embodiment of the present disclosure.
(18) FIG. **7**B is a schematic view of how to operate the blood vessel anastomosis set according to yet another specific embodiment of the present disclosure.
(19) FIG. **7**C is a schematic view of how to operate the blood vessel anastomosis set according to yet another specific embodiment of the present disclosure.
(20) FIG. **7**D is a schematic view of how to operate the blood vessel anastomosis set according to yet another specific embodiment of the present disclosure.
(21) FIG. **7**E is a schematic view of how to operate the blood vessel anastomosis set according to yet another specific embodiment of the present disclosure.
(22) FIG. **7**F is a schematic view of how to operate the blood vessel anastomosis set according to yet another specific embodiment of the present disclosure.
(23) FIG. **8**A is a schematic view of a blood vessel fixation device with a foldable blood vessel widening portion according to a specific embodiment of the present disclosure.
(24) FIG. **8**B is a schematic view of the blood vessel fixation device with the foldable blood vessel widening portion according to a specific embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(25) Referring to FIG. **1**A through FIG. **1**G, there are shown schematic views of how to operate a blood vessel anastomosis set according to a specific embodiment of the present disclosure. In the embodiment illustrated by FIG. **1**A, a blood vessel anastomosis set **100** comprises a blood vessel anastomosis device **110** and a blood vessel fixation device **120**. The blood vessel anastomosis device **110** has anastomosis needles **110**A, **110**B. The blood vessel fixation device **120** further comprises a first blood vessel widening portion **122**, a blood vessel fixation portion **124** and a second blood vessel widening portion **126**. The second blood vessel widening portion **126** is movably received in the blood vessel fixation device **120**. The blood vessel fixation portion **124** has a receiving portion **124**A, in which the receiving portion **124**A is a cavity arranged on the blood vessel fixation portion **124**. The front end of the first blood vessel widening portion **122** has an oblique surface for widening a blood vessel cut gradually. FIG. **1**A serves an exemplary purpose only; consequently, the front end of the first blood vessel widening portion **122** is not necessarily an oblique surface but can be a concave arcuate surface or convex arcuate surface as needed to facilitate the gradual widening of the blood vessel cut.
(26) As shown in FIG. **1**A, the blood vessel anastomosis device **110** of the blood vessel anastomosis set **100** is fitted around a blood vessel cut **910** of a blood vessel **900**. The blood vessel anastomosis device **110** is of a ring-shaped structure; consequently, the blood vessel cut **910** can

penetrate the inner hole of the blood vessel anastomosis device **110**. Referring to FIG. **1**B, after the blood vessel anastomosis device **110** has been fitted around the blood vessel cut **910** of the blood vessel **900**, the front end of the blood vessel fixation device **120** is inserted through the blood vessel cut **910** into the blood vessel **900**, whereas the blood vessel fixation device **120** in the blood vessel **900** penetrates the inner hole of the blood vessel anastomosis device **110**. Then, the second blood vessel widening portion **126** is protruded from the front end of the blood vessel fixation device **120**, such that the second blood vessel widening portion **126** is inserted into the blood vessel **900**. At this point in time, the second blood vessel widening portion **126** spreads and widens the blood vessel **900** from inside. Referring to FIG. **1**C, after the second blood vessel widening portion **126** has widened the blood vessel **900** from inside, the blood vessel fixation device **120** is moved backward (i.e., toward the blood vessel cut **910**), such that the wall of the blood vessel **900** is clamped by and between the second blood vessel widening portion **126** and the blood vessel anastomosis device **110**, thereby allowing the blood vessel **900** to be temporarily fixed in place while an anastomosis procedure is being performed thereon.
(27) Referring to FIG. **1**D, after the blood vessel **900** has been temporarily fixed in place with the second blood vessel widening portion **126**, the first blood vessel widening portion **122** is manipulated in such a manner to cause the front end of the first blood vessel widening portion **122** to enter the blood vessel cut **910** (i.e., move toward the blood vessel cut **910**); consequently, the first blood vessel widening portion **122** widens the blood vessel cut **910** gradually and allows the blood vessel cut **910** to come into contact with the blood vessel fixation portion **124**. Then, the first blood vessel widening portion **122** is manipulated continuously in such a manner to fix the blood vessel cut **910** to the anastomosis needles **110**A, **110**B of the blood vessel anastomosis device **110** (as shown in FIG. **1**E) with the blood vessel fixation portion **124**. At this point in time, the anastomosis needles **110**A, **110**B are received in the receiving portion **124**A. During the anastomosis procedure, the blood vessel fixation device **120** is not in contact with the anastomosis needles **110**A, **110**B to preclude the chance that the anastomosis needles **110**A, **110**B may be compressed by the blood vessel fixation device **120** and thus damaged. In a specific embodiment, the anastomosis needles **110**A, **110**B has barbs whereby the blood vessel cut **910** is fixed to the anastomosis needles **110**A, **110**B firmly.
(28) Referring to FIG. **1**F, after the blood vessel cut **910** has been fixed to the anastomosis needles **110**A, **110**B of the blood vessel anastomosis device **110**, the first blood vessel widening portion **122** is manipulated in such a manner to be moved away from the blood vessel cut **910**. Referring to FIG. **1**G, as soon as the first blood vessel widening portion **122** is moved away from the blood vessel cut **910**, the second blood vessel widening portion **126** is retracted into the blood vessel fixation device **120**, and the blood vessel fixation device **120** is moved out of the blood vessel cut **910**. In a specific embodiment, the step shown in FIG. **1**F is dispensed with, that is, the second blood vessel widening portion **126** is retracted into the blood vessel fixation device **120** directly, and the blood vessel fixation device **120** is moved out of the blood vessel cut **910**, thereby dispensing the need to manipulate the first blood vessel widening portion **122** in such a manner to cause the first blood vessel widening portion **122** to move in a direction away from the blood vessel cut **910**.
(29) Referring to FIG. **2**, there is shown a schematic view of a blood vessel fixation device comprising a second blood vessel widening portion according to a specific embodiment of the present disclosure. In the embodiment illustrated by FIG. **2**, a blood vessel fixation device **220** further comprises retracting portions **228**A, **228**B, in which the retracting portions **228**A, **228**B are humps arranged in the blood vessel fixation device **220**. Each retracting portion is corresponding to a respective blood vessel widening portion, and therefore, as soon as a second blood vessel widening portion **226** is retracted into the blood vessel fixation device **220**, the retracting portions **228**A, **228**B cause the second blood vessel widening portion **226** to restore a retracted state thereof early. Therefore, the retraction of the second blood vessel widening portion **226** into the blood

vessel fixation device **220** precludes the chance that the wall of the blood vessel **900** may be excessively compressed by clamping portions **226**A, **226**B of the second blood vessel widening portion **226** and a blood vessel anastomosis device **210** and thus damaged.

(30) Referring to FIG. **3**, there is a schematic view of a blood vessel anastomosis device according to a specific embodiment of the present disclosure. In the embodiment illustrated by FIG. **3**, a blood vessel anastomosis device **310** is of a ring-shaped structure and has an inner hole **318**. Therefore, the blood vessel can penetrate the inner hole **318**. The blood vessel anastomosis device **310** has anastomosis needles **310**A, **310**B, **310**C and anastomosis needle holes **310**F, **310**G, **310**H. Therefore, after a first blood vessel anastomosis device and a second blood vessel anastomosis device have been fixed to two blood vessel cuts, respectively, the anastomosis needles of the first blood vessel anastomosis device are inserted into the anastomosis needle holes of the second blood vessel anastomosis device, and the anastomosis needles of the second blood vessel anastomosis device are separably inserted into the anastomosis needle holes of the first blood vessel anastomosis device, thereby connecting the two blood vessel cuts to each other.

(31) Referring to FIG. **4**, there is shown a schematic view of a blood vessel anastomosis device according to another specific embodiment of the present disclosure. In the embodiment illustrated by FIG. **4**, a blood vessel anastomosis device **410** comprises a front-end anastomosis portion **412** and a rear-end anastomosis portion **414**. The rear-end anastomosis portion **414** has anastomosis needles **414**A, **414**B, **414**C. The anastomosis needles **414**A, **414**B, **414**C penetrate the front-end anastomosis portion **412**, such that the rear-end anastomosis portion **414** is movably connected to the front-end anastomosis portion **412** through the anastomosis needles **414**A, **414**B, **414**C. In a specific embodiment, the front-end anastomosis portion **412** and the rear-end anastomosis portion **414** each have anastomosis needle holes. In another specific embodiment, only the front-end anastomosis portion **412** has anastomosis needle holes. Therefore, the first blood vessel anastomosis device and the second blood vessel anastomosis device, which are fixed to different blood vessel cuts, respectively, are connected to each other through the anastomosis needles and the anastomosis needle holes.

(32) Referring to FIG. **5**A through FIG. **5**E, there are shown schematic views of how to operate the blood vessel anastomosis set according to another specific embodiment of the present disclosure. In the embodiment illustrated by FIG. **5**A, a blood vessel anastomosis set **500** comprises a blood vessel anastomosis device **510** and a blood vessel fixation device **520**. The blood vessel anastomosis device **510** further comprises a front-end anastomosis portion **512** and a rear-end anastomosis portion **514**. The blood vessel fixation device **520** further comprises a blood vessel widening portion **522** and a blood vessel fixation portion **524**. The blood vessel fixation portion **524** is separably disposed at the blood vessel fixation device **520**. In this specific embodiment, the rear-end anastomosis portion **514** has anastomosis needles **514**A, **514**B. The anastomosis needles **514**A, **514**B penetrate the front-end anastomosis portion **512**, such that the rear-end anastomosis portion **514** is movably connected to the front-end anastomosis portion **512** through the anastomosis needles **514**A, **514**B. The front-end anastomosis portion **512** of the blood vessel anastomosis device **510** has a snap-engagement portion **512**A. The blood vessel fixation portion **524** is snap-engaged with the snap-engagement portion **512**A.

(33) Operation of the blood vessel anastomosis set according to another specific embodiment illustrated by FIG. **5**A through FIG. **5**E is described below. Referring to FIG. **5**A, the blood vessel anastomosis device **510** of the blood vessel anastomosis set **500** is fitted around the blood vessel cut **910** of the blood vessel **900**. The blood vessel anastomosis device **510** is of a ring-shaped structure; consequently, the blood vessel cut **910** penetrates the inner hole of the blood vessel anastomosis device **510**. Referring to FIG. **5**B, after the blood vessel anastomosis device **510** has been fitted around the blood vessel cut **910** of the blood vessel **900**, the blood vessel fixation device **520** is manipulated in such a manner to cause the front end of the blood vessel widening portion **522** to enter the blood vessel cut **910** (i.e., move toward the blood vessel cut **910**); consequently,

the blood vessel widening portion **522** widens the blood vessel cut **910** gradually and allows the blood vessel cut **910** to come into contact with the blood vessel fixation portion **524**.

(34) Referring to FIG. **5**C, after the blood vessel cut **910** has come into contact with the blood vessel fixation portion **524**, the blood vessel fixation device **520** is manipulated continuously so as for the blood vessel fixation portion **524** to be snap-engaged with the snap-engagement portion **512**A of the front-end anastomosis portion **512** and for the blood vessel cut **910** to be clamped between the blood vessel fixation portion **524** and the front-end anastomosis portion **512** of the blood vessel anastomosis device **510**. Consequently, the blood vessel fixation portion **524** can fix the blood vessel cut **910** to the blood vessel anastomosis device **510**. After the blood vessel fixation portion **524** has fixed the blood vessel cut **910** to the blood vessel anastomosis device **510**, the anastomosis needles **514**A, **514**B do not protrude from the front-end anastomosis portion **512**. Therefore, the anastomosis needles **514**A, **514**B are unlikely to be compressed by the blood vessel fixation device **520** and damaged.

(35) In a specific embodiment, the blood vessel widening portion **522** is made of a deformable material. Therefore, during the anastomosis procedure, the blood vessel widening portion **522** corresponds in shape to the blood vessel anastomosis device **510** and thus pushes the blood vessel cut **910** toward the blood vessel widening portion **522**.

(36) Referring to FIG. **5**D, after blood vessel fixation portion **524** has fixed the blood vessel cut **910** to the blood vessel anastomosis device **510**, the blood vessel fixation device **520** is removed. Then, the rear-end anastomosis portion **514** is pushed toward the front-end anastomosis portion **512**; consequently, the rear-end anastomosis portion **514** abuts against the front-end anastomosis portion **512**, and the anastomosis needles **514**A, **514**B penetrate the blood vessel cut **910** to thereby protrude from the front-end anastomosis portion **512** (as shown in FIG. **5**E). Therefore, the blood vessel cut **910** is fixed to the blood vessel anastomosis device **510** firmly.

(37) Referring to FIG. **6**, there is shown a schematic view of a blood vessel fixation portion is snap-engaged with a blood vessel anastomosis device according to a specific embodiment of the present disclosure. In the embodiment illustrated by FIG. **6**, upon fixation of the blood vessel cut **910** to a blood vessel anastomosis device **610**, penetration of anastomosis needles **614**A, **614**B into the blood vessel cut **910** and protrusion of the anastomosis needles **614**A, **614**B from a front-end anastomosis portion **612**, the anastomosis needles **614**A, **614**B protrude from a blood vessel fixation portion **624**. Therefore, as soon as a first blood vessel anastomosis device and a second blood vessel anastomosis device are fixed to two blood vessel cuts, respectively, the two blood vessel cuts are connected to each other through the anastomosis needles of the first blood vessel anastomosis device and the anastomosis needles of the second blood vessel anastomosis device.

(38) Referring to FIG. **7**A through FIG. **7**F, there are shown schematic views of how to operate the blood vessel anastomosis set according to yet another specific embodiment of the present disclosure. In the embodiment illustrated by FIG. **7**A, the blood vessel anastomosis set **700** comprises a blood vessel anastomosis device **710** and a blood vessel fixation device **720**. The blood vessel anastomosis device **710** further comprises a front-end anastomosis portion **712** and a rear-end anastomosis portion **714**. The rear-end anastomosis portion **714** has anastomosis needles **714**A, **714**B. The anastomosis needles **714**A, **714**B penetrate the front-end anastomosis portion **712**, such that the rear-end anastomosis portion **714** is movably connected to the front-end anastomosis portion **712** through the anastomosis needles **714**A, **714**B.

(39) The anastomosis procedure performed according to the specific embodiment illustrated by FIG. **7**A through FIG. **7**F is described below. Referring to FIG. **7**A, the blood vessel anastomosis device **710** of the blood vessel anastomosis set **700** is fitted around the blood vessel cut **910** of the blood vessel **900**. The blood vessel anastomosis device **710** is of a ring-shaped structure; consequently, the blood vessel cut **910** penetrates the inner hole of the blood vessel anastomosis device **710**. Referring to FIG. **7**B, as soon as the blood vessel anastomosis device **710** is fitted around the blood vessel cut **910**, the blood vessel fixation device **720** enters the blood vessel cut

910 by an entering angle.

(40) Referring to FIG. **7**C, after the front end of the blood vessel fixation device **720** has entered the blood vessel cut **910**, the blood vessel fixation device **720** is manipulated to adjust its entering angle, such that the front end of the blood vessel fixation device **720** widens the blood vessel cut **910**. Then, the blood vessel fixation device **720** is manipulated to adjust its entering angle, such that the widened blood vessel cut **910** is fitted to the blood vessel anastomosis device **710** (as shown in FIG. **7**D). At this point in time, instead of removing blood vessel fixation device **720**, it is necessary to manipulate the blood vessel fixation device **720** in such a manner to press the blood vessel cut **910** slightly and thus prevent the blood vessel cut **910** from slipping out of the blood vessel anastomosis device **710**.

(41) Referring to FIG. **7**E, after the widened blood vessel cut **910** has been fitted to the blood vessel anastomosis device **710**, the rear-end anastomosis portion **714** is pushed toward the front-end anastomosis portion **712**; consequently, the rear-end anastomosis portion **714** abuts against the front-end anastomosis portion **712**, and the anastomosis needles **714**A, **714**B penetrate the blood vessel cut **910**. Therefore, the blood vessel cut **910** is fixed to the anastomosis needles **714**A, **714**B of the blood vessel anastomosis device **710**. Then, the blood vessel fixation device **720** is removed from the blood vessel cut **910** (as shown in FIG. **7**F). After the blood vessel fixation portion **724** has fixed the blood vessel cut **910** to the blood vessel anastomosis device **710**, the anastomosis needles **714**A, **714**B do not protrude from the front-end anastomosis portion **712**. Therefore, the anastomosis needles **714**A, **714**B are unlikely to be compressed by the blood vessel fixation device **720** and damaged. As soon as a first blood vessel anastomosis device and a second blood vessel anastomosis device are fixed to two blood vessel cuts, respectively, the two blood vessel cuts are connected to each other through the anastomosis needles of the first blood vessel anastomosis device and the anastomosis needles of the second blood vessel anastomosis device.

(42) Referring to FIG. **8**A, there is shown a schematic view of a blood vessel fixation device with a foldable blood vessel widening portion according to a specific embodiment of the present disclosure. In the embodiment illustrated by FIG. **8**A, a foldable blood vessel widening portion **829** of a blood vessel fixation device **820** is folded, and thus the foldable blood vessel widening portion **829** enters the blood vessel cut readily.

(43) Referring to FIG. **8**B, there is shown a schematic view of the blood vessel fixation device with the foldable blood vessel widening portion according to a specific embodiment of the present disclosure. In the embodiment illustrated by FIG. **8**B, the foldable blood vessel widening portion **829** of the blood vessel fixation device **820** spreads. After the foldable blood vessel widening portion **829** has entered the blood vessel cut, the blood vessel fixation device **820** is manipulated in such a manner to cause the foldable blood vessel widening portion **829** to spread. Therefore, the blood vessel cut is widened with the foldable blood vessel widening portion **829**, and then the widened blood vessel cut is fitted to the blood vessel anastomosis device.

(44) The blood vessel anastomosis set of the present disclosure is also suitable for performing various anastomosis procedures on colons, small intestines, ureters, bile ducts, etc.

(45) The blood vessel anastomosis set of the present disclosure is depicted by the accompanying drawings and described above. The specific embodiments of the present disclosure serve an exemplary purpose only. Various changes can be made to the specific embodiments of the present disclosure without departing from the claims of the present disclosure and spirit thereof and deemed falling within the scope of the claims of the present disclosure. Therefore, the specific embodiments described herein are not restrictive of the present disclosure; thus, the actual scope and spirit of the present disclosure is defined by the appended claims.

Claims

1. A blood vessel anastomosis set for anastomosing a blood vessel cut of a blood vessel, the blood vessel anastomosis set comprising: a blood vessel anastomosis device, comprising: a front-end anastomosis portion; and a rear-end anastomosis portion having a plurality of anastomosis needles, wherein the plurality of anastomosis needles penetrate the front-end anastomosis portion, such that the rear-end anastomosis portion is movably connected to the front-end anastomosis portion through the anastomosis needles; and a blood vessel fixation device for widening the blood vessel cut to allow the blood vessel cut to be fixed to the blood vessel anastomosis device; wherein the blood vessel fixation device comprises a blood vessel widening portion and a blood vessel fixation portion; wherein the first blood vessel widening portion is configured to widen the blood vessel cut gradually and allows the blood vessel cut to come into contact with the blood vessel fixation portion, and thereby cause the blood vessel fixation portion fixing the blood vessel cut to the blood vessel anastomosis device; wherein the blood vessel anastomosis device is connected through the plurality of anastomosis needles to a second blood vessel anastomosis device, wherein the second blood vessel anastomosis device is configured to fix to another blood vessel cut; wherein the blood vessel anastomosis device has a snap-engagement portion, and the blood vessel fixation portion is disposed at the blood vessel fixation device, in which the blood vessel fixation portion is separable with the blood vessel fixation device, wherein, as soon as the blood vessel fixation portion fixes the blood vessel cut to the blood vessel anastomosis device, the blood vessel fixation portion is snap-engaged with the snap-engagement portion, and the blood vessel cut is clamped between the blood vessel fixation portion and the blood vessel anastomosis device.

2. The blood vessel anastomosis set of claim 1, wherein the plurality of anastomosis needles do not protrude from the front-end anastomosis portion when the blood vessel fixation device fixes the blood vessel cut to the blood vessel anastomosis device.